Analysis and Enhancement of the Flexibility Potential of a Swiss Medium-Voltage Distribution Grid

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Article On the Flexibility Potential of a Swiss Distribution Grid: Opportunities and Limitations

1 Introduction

The increasing deployment of distributed energy resources (DERs), such as photovoltaic systems, battery energy storage systems, electric vehicles, and flexible loads, enables distribution grids to provide flexibility to the upper-level grid. However, device-level flexibility cannot always be delivered at the Point of Common Coupling (PCC), as it may be restricted by voltage and line-loading limits, grid topology, baseline conditions, and DER location.

This project analyses the Feasible Flexibility Operating Region (FFOR) of a synthetic 20 kV medium-voltage grid in Eschenbach SG, Switzerland. The grid consists of 51 buses, 50 lines, and 26 load nodes, and is evaluated under future scenarios for 2030, 2040, and 2050.

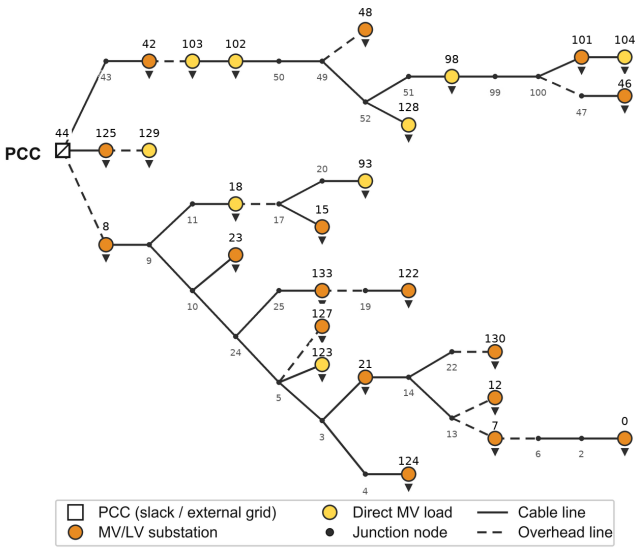


Figure 1: Topology of the synthetic medium-voltage distribution grid.

The objective is to identify the factors limiting the FFOR and assess three flexibility-enhancement strategies: critical-line reinforcement to increase transfer capacity, additional BESS installation to provide energy-constrained bidirectional active and reactive flexibility, and large-scale PV integration to increase downward active-power flexibility through curtailment while reducing branch loading and congestion. The study addresses the following question:

How do network constraints, future DER penetration, resource location, and flexibility activation duration shape the FFOR of a medium-voltage distribution grid, and under which conditions are line reinforcement, BESS installation, and PV integration most effective?

2 Methodology

First, the network model and operational baselines are defined. The grid topology is taken from [1], and DERs are allocated to the grid nodes using the dataset developed in [2]. Following the aggregation procedure described in Appendix F, its LV quantities are mapped to the corresponding MV nodes to obtain nodal profiles of conventional demand, EV charging, PV generation, HP consumption, and BESS capacity for 2030, 2040, and 2050. The dataset projections account for the spatial and socioeconomic evolution of DER deployment. The operational feasibility of each baseline is assessed using the linearized power-flow model described in Appendix D, considering both line apparent-power limits and bus-voltage limits. The FFOR is then computed through a multi-timestep linear optimization model implemented in Python and solved with Gurobi. Following [3], the region is initialized from four cardinal solutions and iteratively refined along the outward normals of its facets until the marginal area increase falls below a prescribed tolerance.

Line Reinforcement The multi-timestep model was configured to compute an instantaneous single-timestep FFOR and identify the line and voltage constraints binding at each boundary point. The analysis was performed on the 15th day of each month, at 12:00 and 19:00, for 2030, 2040, and 2050. The most frequently binding lines were selected for reinforcement using the grid data and conductor catalogue from [1]. In the synthetic grid, all overhead lines (OHLs) share one conductor type and all cables another, each with common electrical parameters and nominal apparent-power limit S_{nom}. OHLs of type 48-AL1/8-ST1A were replaced by conductors with a cross-section of 70 mm^2 instead of 48 mm^2, reducing resistance by 30.4% and reactance by 3.2%, while increasing S_{nom} by 38.1%; the small change in shunt capacitance was neglected. Cables of type NA2XS2Y 1x70 RM/25 12 were replaced by conductors with a cross-section of 150 mm^2 instead of 70 mm^2, reducing resistance by 53.5% and reactance by 16.7%, increasing capacitance by 89.5%, and raising S_{nom} by 43.2%. The capacitance variation was retained for cables because of its greater significance. The multi-timestep model was then used to quantify the resulting flexibility increase.

BESS Placement and Sizing Sensitivity Battery energy storage systems are modelled as grid-side controllable DERs at selected MV buses[2]. For each candidate placement or rating, the FFOR at the PCC is recomputed and the impact is quantified through the area increase $\Delta A = A_{\mathrm{BESS}} - A_{\mathrm{baseline}}$. The BESS provides P-Q controllability within an apparent-power limit and is coupled across timesteps through state-of-charge dynamics ($\mathrm{SoC}_0 = 0.5, \mathrm{SoC} \in [0.2, 0.8]$). A single signed power variable with unit efficiency preserves LP linearity and yields an upper-bound estimate without altering the ranking. The analysis proceeds in three steps. A 1 MW / 2.5 MWh unit is first swept over all MV buses (excluding the PCC) to identify high-value locations; node 129 is evaluated across months and DER scenarios (2030, 2040, 2050) [2]; finally, the rating is varied at fixed $E/P = 2.5$ h across the same scenarios. Sustained horizons of $N \in \{1, 4\}$ at $dt = 1$ h are considered, capturing both instantaneous and energy-limited storage effects.

PV Placement Sensitivity This analysis evaluates large-scale PV placement in two phases. First, PV fields of 1–3 MW are simulated at every grid node on the 15th of each month at 12:00, using 1–4 h sustained horizons across the 2030, 2040, and 2050 DER configurations [2]. Second, the nodes providing the greatest FFOR-area and downward-flexibility improvements are further analyzed using monthly representative days for the same future scenarios. Representative days are selected by comparing each day's fixed-demand, heat-pump, EV, and distributed-PV profiles with their corresponding monthly median profiles. The absolute deviations are normalized, summed into a total deviation score, and the day with the lowest score is selected. This phase applies the same capacity and sustained-horizon sensitivities to both individual nodes and coupled installations, for which the total PV capacity is divided equally between the two selected locations. Large-scale PV timeseries are generated by normalizing the local distributed-PV profile by its yearly peak and multiplying it by the installed capacity. Where no local profile is available, the average distributed-PV production profile is scaled identically.

3 Results

Line Reinforcement The baseline feasibility analysis identified 649 infeasible operating points in 2040 and 2050, particularly during the evening peak at 19:00, all caused by line apparent-power violations. Voltage limits were never binding. Based on these violations and on the FFOR binding-constraint analysis, lines 8–44, 42–43, 42–103, 8–9, 43–44, 10–24, and 9–10, shown in Figure 1, were reinforced. Figure 2 summarizes the main effects on the FFOR. In 2030, line reinforcement generally provides only limited benefits because the region is mainly constrained by the relatively small amount of flexible capacity. The impact is larger in 2050, when higher DER penetration increases the theoretical flexibility but higher HP and EV demand brings the baseline operation closer to the line limits. Reinforcement therefore allows a larger share of device-level flexibility to be delivered at the PCC. Figure 3 confirms that reinforcement mainly increases negative active-power flexibility, corresponding to increased consumption or reduced generation. These actions

raise the loading of the critical branches and are therefore constrained by line limits. By contrast, positive flexibility, corresponding to increased generation or reduced consumption, generally reduces branch loading and remains mainly limited by DER capacity. Consistently, the previous binding-constraint analysis shows that boundary points in the positive active-power flexibility region are generally determined by DER saturation rather than by line constraints. More generally, negative flexibility is larger because PV systems can reduce their generation through curtailment but cannot increase it above the available power. The reinforcement benefit also varies with time of day and season. At 12:00, it is smaller in summer because low HP consumption leaves the network less heavily loaded, while PV curtailment tends to relieve rather than increase line loading. In the other months, higher demand makes line constraints more relevant. At 20:00, the baseline is more heavily loaded, especially in winter, but the available flexibility is smaller; consequently, the gain remains limited in 2030 and 2040 and becomes substantial mainly in 2050.

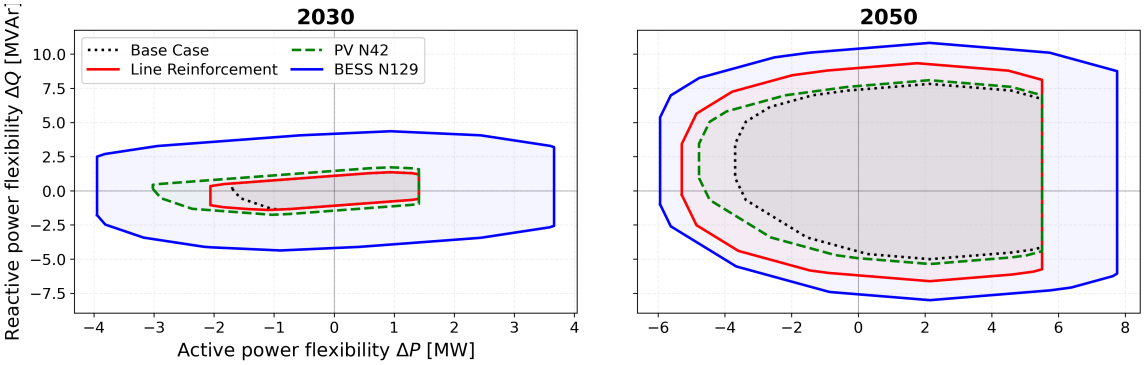


Figure 2: FFOR boundary comparison on 15 December at 12:00 for 2030 (left) and 2050 (right), with 1-hour sustained flexibility.

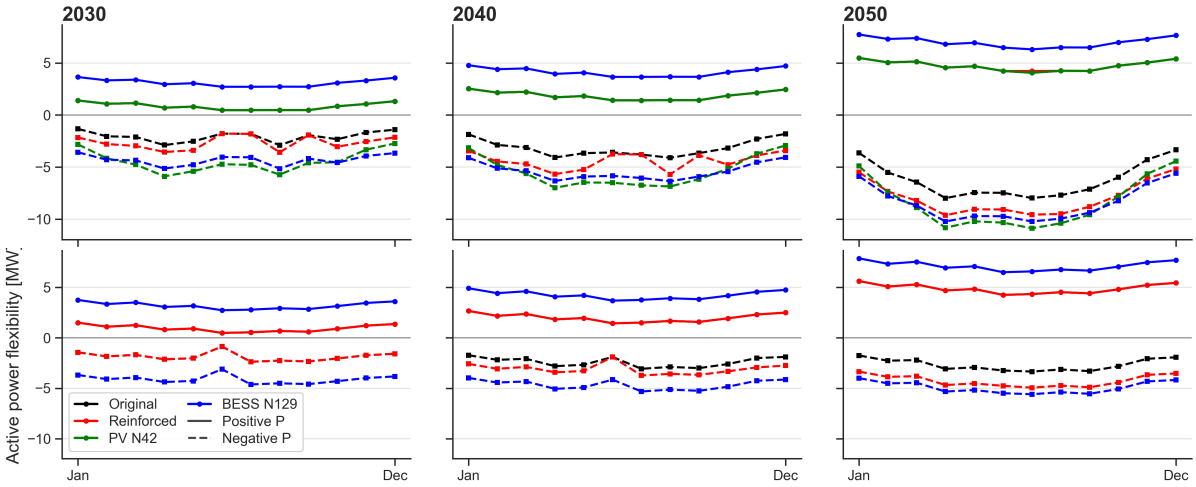


Figure 3: Monthly active-power flexibility across representative days in 2030, 2040, and 2050, evaluated at 12:00 (top) and 20:00 (bottom), with flexibility sustained for one hour.

BESS Placement and Sizing Sensitivity A 1 MW / 2.5 MWh auxiliary BESS is modelled as a grid-side unit connected to each candidate MV bus, excluding the PCC. For each placement, the 2050 FFOR at the PCC is recomputed on 15 December at 12:00 with $N = 4$ and $dt = 1$ h, and the benefit is measured as $\Delta A = A_{\mathrm{BESS}} - A_{\mathrm{baseline}}$. The location screening shows a strong spatial dependence of the BESS contribution, with gains ranging from $+1.05$ to $+18.08$ MW·MVAr. The highest gains occur on the stub feeder, where nodes 125 and 129 reach about $+18.08$ MW·MVAr ($+34.8\%$); the upper feeder gives intermediate gains, with mean $\Delta A = +10.78$ MW·MVAr, while the lower feeder and remaining nodes are much lower, with mean $\Delta A = +3.70$ MW·MVAr.

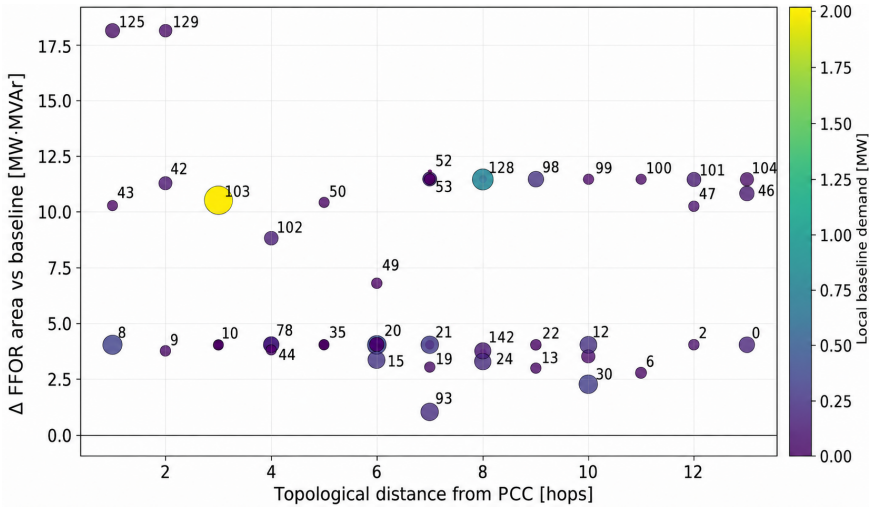


Figure 4: BESS area gain ΔA as a function of topological distance from the PCC. Marker labels indicate node osmids. Colour and marker size represent local baseline demand.

Line 8–44, at the lower-feeder head, reaches 100% loading at 14 boundary vertices and already carries 72.5% of its thermal limit; line 103–42 binds at the remaining line-constrained vertices. In contrast, nodes 125 and 129 lie on a lightly loaded MV stub, where the aggregated LV demand is small relative to the MV branch rating. Their larger gains reflect available thermal headroom in the MV-equivalent model, where LV cables and local transformer limits are not explicitly represented.

Seasonal, scenario and size effects. At node 129, ΔA is positive in all tested months and years. Mean monthly ΔA grows $3.7\times$ from 2030 to 2050 (6.0 to 22.0 MW·MVAr), and the seasonal range widens from 3.96–7.69 to 17.11–31.44 MW·MVAr: the same unit becomes both more valuable and more sensitive to the operating condition as DER penetration, electrified demand and active network constraints grow. At fixed $E/P = 2.5$ h, raising the BESS from 1 to 3 MW yields a superlinear response, e.g. ΔA from 17.1 to 54.0 MW·MVAr in 2050 ($+215\%$): each additional MW contributes more than the previous one, since the FFOR boundary expands along directions where the network still has thermal headroom. Extending the 2050 sweep to 0.5–4.0 MW gives $\Delta A = 8.49$–70.88 MW·MVAr with no global saturation: line-binding shapes local FFOR directions but does not cap the total area, so at node 129 and within the tested range the storage value is set primarily by the installed capacity rather than by the surrounding network constraints.

PV Sensitivity Analysis The all-nodes screening identifies high-performing locations among both central and peripheral nodes. The small differences between their average gains, reported in Table 2, indicate that distance from the PCC alone does not determine PV-placement performance. Instead, performance depends primarily on local topology and network constraints.

Table 1: Average FFOR-area and downward-flexibility gains.

		Area gain (%)			Negative-P gain (MW)		
Year	Horizon	1 MW	2 MW	3 MW	1 MW	2 MW	3 MW
2030	1 h	56.18	119.91	190.64	1.00	2.00	2.98
	4 h	38.87	68.07	93.24	0.70	1.18	1.52
2040	1 h	27.48	55.49	83.23	1.00	2.00	3.00
	4 h	26.28	43.49	55.97	0.72	1.25	1.66
2050	1 h	8.49	15.60	21.21	0.94	1.87	2.78
	4 h	5.37	8.99	10.62	0.35	0.63	0.82

Table 2: Line 8–44 and nodes.

	Binding scenarios (%)			
PV	2030	2040	2050	All
0 MW	66.7	100.0	100.0	88.5
1 MW	59.7	99.5	100.0	85.9
2 MW	48.4	97.6	100.0	81.4
3 MW	37.1	99.1	100.0	78.0

Central–peripheral PV3 comparison	
Metric	Maximum $\lvert C - P \rvert$
FFOR-area gain	3.1 pp
Negative-P gain	0.07 MW

Table 1 compares each feasible PV scenario with its matched baseline (PV0). For H1, the downward-flexibility gain scales almost proportionally with installed PV capacity. Longer sustained durations reduce both downward-flexibility and FFOR-area gains because of inter-temporal constraints. Table 2 shows that additional PV can relieve Line 8–44 in 2030, but has limited influence in the more

constrained 2040 and 2050 systems. Overall, large-scale PV expands the FFOR and can relieve specific bottlenecks under less stressed conditions, but it cannot replace structural reinforcement when future network constraints become dominant.

Nodes Analysis Although rankings vary across grid states, Node 42 provides the most consistent FFOR-area performance, ranking first in 2030 and 2050, while remaining competitive in downward flexibility.

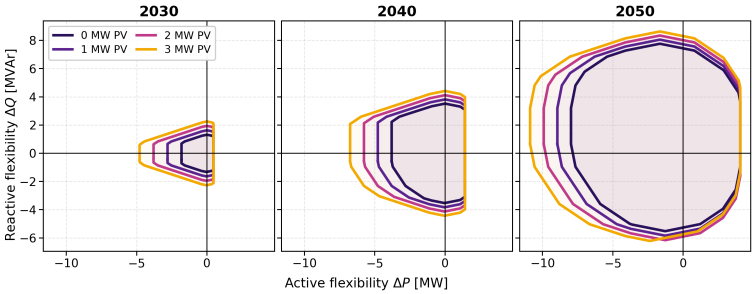


Figure 5: Node 42 July representative-day (24th) at 12.00 p.m., H1 FFORs across 0–3 MW of PV installation.

Figure 5 shows that additional PV leaves the positive boundary almost unchanged. At H1, nearly the entire installed PV capacity is available as downward flexibility through curtailment. Also, both reactive-power boundaries expand, increasing the overall FFOR area, with greater positive reactive flexibility in 2050.

The binding limits evolve over time: device constraints (i.e., DER capacities) delimit the 2030 FFOR, while Line 8–44 and other line-flow constraints increasingly restrict the 2040 and 2050 PQ space. Consequently, PV remains effective for downward flexibility, but network congestion progressively limits its reactive-power contribution, resulting in a more asymmetric FFOR. Coupled-placement analysis shows that downward flexibility gains are similar across placements, while positive active flexibility depends strongly on location. Pair 42+46 and single node 42 placements preserve the baseline positive boundary and July 2050 feasibility, unlike single node 46, which is restricted by Line 47–100. Pair 9+49 suffers the greatest positive-boundary reduction due to Line 8–44. Finally, coupled deployment mitigates local constraints, but provides no intrinsic flexibility synergy over single-node placement.

4 Conclusions

This study shows that increasing DER penetration does not automatically translate into greater flexibility at the PCC. From 2030 to 2050, device-level flexibility increases, but rising EV and HP demand moves the network closer to line apparent-power limits, restricting the amount of flexibility that can be delivered. The 649 infeasible baseline operating points identified in 2040 and 2050 indicate that network interventions will become necessary under future electrification. The three enhancement strategies address different constraints. Line reinforcement expands the FFOR, primarily in the negative active-power direction, and is the only strategy evaluated for restoring baseline feasibility across the full 24-hour period. BESS appears to be the most effective measure for expanding the FFOR, especially for positive active power and reactive power flexibility, but its effectiveness depends strongly on placement and on installed power and energy capacity. However, whether BESS can restore infeasible baseline operating points was not evaluated. In baseline calculations, batteries were assumed inactive, with zero active and reactive power because corrective dispatch would require modelling a consistent prior state-of-charge trajectory. Large-scale PV mainly increases negative active-power flexibility through curtailment and can relieve daytime congestion, but its contribution decreases during evening peaks and under highly constrained conditions. The appropriate intervention therefore depends on the objective, operating point, and location of binding constraints. Line reinforcement addresses structural bottlenecks, BESS provides apparently superior but energy-limited flexibility, and PV provides daytime curtailment-based flexibility. A key limitation is that PV and BESS were evaluated independently, with the initial BESS state of charge fixed across simulations. Coordinated PV–BESS operation could store daytime PV production and shift flexibility toward later high-load periods, but the analysis cannot determine whether this would reduce the need for line reinforcement, and it is left as future work.

Appendix A: Index Sets

$\mathcal{N}$: Set of MV nodes in the distribution grid
$0 \in \mathcal{N}$: Slack node, i.e., the Point of Common Coupling (PCC)
$\mathcal{N}_i$: Set of nodes directly connected to node i
$\mathcal{L}$: Set of directed MV lines, indexed by (i, j)
$\mathcal{T}$: Set of timesteps in the multi-timestep flexibility horizon
$\mathcal{H} \subseteq \mathcal{N}$: Set of nodes with aggregated heat-pump load
$\mathcal{B} \subseteq \mathcal{N}$: Set of nodes with distributed battery energy storage systems
$\mathcal{B}_{\text{ls}} \subseteq \mathcal{N}$: Set of nodes with large-scale battery energy storage systems
$\mathcal{P} \subseteq \mathcal{N}$: Set of nodes with photovoltaic generation
$\mathcal{K}$: Set of linear facets used to approximate circular apparent-power limits
$\mathcal{D}$: Set of search directions in the PQ-plane used to construct the FFOR.

The set $\mathcal{T} = \{0, \dots, H\}$ contains the operating points considered during an H-interval flexibility activation; for a single-interval activation, $\mathcal{T} = \{0, 1\}$.

Appendix B: Decision Variables

$P^{\text{flex}}_{\text{PCC}}, Q^{\text{flex}}_{\text{PCC}} \in \mathbb{R}$: Sustained active and reactive flexibility at the PCC [p.u.]
$P_{0t}, Q_{0t} \in \mathbb{R}$: Active and reactive power exchange at the PCC at time t [p.u.]
$p_{it}, q_{it} \in \mathbb{R}$: Net active and reactive injection at node i and time t [kW, kvar]
$\theta_{it} \in \mathbb{R}$: Voltage angle at node i and time t [rad]
$\Delta U_{it} \in \mathbb{R}$: Voltage magnitude deviation from 1 p.u. at node i and time t [p.u.]
$P^{\ell}_{ijt}, Q^{\ell}_{ijt} \in \mathbb{R}$: Active and reactive power flow on line (i, j) at time t [p.u.]
$P^{\text{PV}}_{it} \in \mathbb{R}_{\geq 0}, Q^{\text{PV}}_{it} \in \mathbb{R}$: Active and reactive PV injection at node i and time t [kW, kvar]
$P^{\text{HP}}_{it} \in \mathbb{R}_{\geq 0}, Q^{\text{HP}}_{it} \in \mathbb{R}$: Active and reactive HP consumption at node i and time t [kW, kvar]
$T^{\text{in}}_{it} \in \mathbb{R}_{\geq 0}$: Indoor temperature state of the aggregated HP load at node i [°C]
$P^{\text{B}}_{it}, Q^{\text{B}}_{it} \in \mathbb{R}$: Active and reactive BESS injection at node i and time t [kW, kvar]
$\text{SOC}_{it} \in \mathbb{R}_{\geq 0}$: State of charge of the BESS at node i and time t [-].

The sign convention is generation-positive for nodal injections. Hence, PV and BESS discharge enter the nodal balance as positive injections, whereas conventional demand, EV charging and heat-pump consumption enter as negative injections. For the BESS variable, $P^{\text{B}}_{it} > 0$ denotes discharge and $P^{\text{B}}_{it} < 0$ denotes charging. The PCC exchange uses an import-positive convention, such that $P_{0t} > 0$ and $Q_{0t} > 0$ denote power imported from the upstream grid.

Appendix C: Parameters

Model parameters

$P_{0t}^{\text{base}}, Q_{0t}^{\text{base}}$	Baseline active and reactive power exchange at the PCC [kW, kvar].
$P_{it}^{\text{D}}, Q_{it}^{\text{D}}$	Non-controllable demand at node i and time t [kW, kvar].
$P_{it}^{\text{EV}}, Q_{it}^{\text{EV}}$	Baseline EV charging demand at node i and time t [kW, kvar].
$\overline{P}_{it}^{\text{PV}}$	Available PV active power at node i and time t [kW].
$P_{it}^{\text{HP,base}}$	Baseline HP active-power consumption at node i and time t [kW].
$P_{i}^{\text{HP,peak}}$	Maximum heat-pump active-power consumption observed in the annual baseline profile at node i [kW].
α_{HP}	Conservative margin factor used to estimate the maximum flexible HP power from annual peaks. It is set to 0.90 based on consistency checks; for example, at node 122 the 2030 annual peak occurs at -10.0°C, while the scaled value corresponds to an equivalent outdoor temperature of -12.1°C, considered a reasonable severe winter design condition.
$\overline{P}_{i}^{\text{HP}}$	Maximum flexible active-power consumption of the heat-pump asset at node i [kW], defined as $$\overline{P}_{i}^{\text{HP}} = \frac{P_{i}^{\text{HP,peak}}}{\alpha_{\text{HP}}}.$$
H_i	Aggregated building thermal conductance at node i [kW/K].
C_i	Aggregated building thermal capacitance at node i [kWh/K].
β_{it}	Heat-pump coefficient of performance at node i and time t.
T_{it}^{out}	Outdoor temperature assigned to node i and time t [°C].
$T^{\text{set}}, \underline{T}, \overline{T}$	Indoor temperature setpoint (20°C, based on [4]), lower comfort bound (19°C), and upper comfort bound (21°C). The ± 1°C band provides limited thermal flexibility while preserving residential comfort and avoiding excessive heating consumption.
$\overline{S}_{i}^{\text{B}}$	BESS apparent-power rating at node i [kVA].
E_{i}^{B}	BESS energy capacity at node i [kWh].
SOC_{i}^{0}	Initial BESS state of charge [-]. Set to 0.60 for distributed BESS and to 0.50 for the auxiliary large-scale BESS, corresponding to the midpoints of their respective admissible SOC ranges. These values represent the pre-flexibility storage condition prior to the optimization horizon.

$\underline{SOC}_i^{\text{dist}}, \overline{SOC}_i^{\text{dist}}$	Minimum and maximum admissible state of charge for distributed BESS [-]. Set to 0.40 and 0.80, respectively, following the SOC operating range used for BESS participation in distribution-grid flexibility in [5]. This conservative window limits deep cycling while preserving energy availability for the end-user.
$\underline{SOC}_i^{\text{ls}}, \overline{SOC}_i^{\text{ls}}$	Minimum and maximum admissible state of charge for the auxiliary large-scale BESS [-]. Set to 0.20 and 0.80, respectively. The inter-temporal SOC formulation follows the BESS modelling approach in [5], while the wider lower bound is introduced here as a modelling assumption to represent deeper cycling capability for a grid-support storage asset.
$\overline{S}_{ij}$	Apparent-power limit of line (i, j) [kVA]. Set to the nominal rating S_{ij}^{nom} from [1], avoiding the use of emergency trip limits for routine flexibility provision.
$\underline{U}, \overline{U}$	Minimum and maximum nodal voltage magnitude [p.u.]. Set to 0.95 and 1.05 p.u. ($\pm 5\%$). This conservative margin ensures the FFOR supports the grid without actively exploring voltage limits (EN 50160 allows $\pm 10\%$ for MV grids [6]), consistent with [5].
Δt	Duration of one timestep [h]. Set to 1 h to match the resolution of the input demand and generation profiles.
$\text{pf}^{\text{PV}}, \text{pf}^{\text{HP}}, \text{pf}^{\text{EV}}$	Power factors used for PV, HP and EV reactive-power modelling. Fixed to 0.95, approximating device feasibility regions from [5] as detailed inverter capabilities are unavailable.
α_d, β_d	Active and reactive weights defining search direction $d \in \mathcal{D}$.
$\lvert\mathcal{K}\rvert$	Number of linear facets used to approximate circular apparent-power limits. Set to 16, representing a reasonable trade-off between geometric accuracy and computational complexity.
r_{ij}, x_{ij}	Series resistance and reactance of line (i, j) [p.u.].
g_{ij}, b_{ij}	Series conductance and susceptance of line (i, j) [p.u.].
b_{ij}^{sh}	Half-line shunt susceptance of line (i, j) [p.u.].
y_{ij}	Series admittance of line (i, j), defined as $y_{ij} = g_{ij} + \text{j}b_{ij} = \frac{1}{r_{ij} + \text{j}x_{ij}}$.
S_{base}	Per unit system base [kVA].
U_{base}	Per unit system base [kV].

Appendix D: Constraints

PCC flexibility definition. The FFOR describes deviations from the baseline power exchange at the PCC. For every timestep in the sustained activation horizon,

$$P_{\mathrm{PCC}}^{\mathrm{flex}} = \frac{P_{0t}^{\mathrm{base}}}{S_{\mathrm{base}}} - P_{0t}, \qquad \forall t \in \mathcal{T},$$
$$Q_{\mathrm{PCC}}^{\mathrm{flex}} = \frac{Q_{0t}^{\mathrm{base}}}{S_{\mathrm{base}}} - Q_{0t}, \qquad \forall t \in \mathcal{T}.$$

Positive active-power flexibility denotes a reduction in the active-power import from the upstream grid relative to the baseline. Equivalently, it represents an increase in the net generation-positive injection of the distribution network. Positive reactive-power flexibility is defined analogously as a reduction in reactive-power import relative to the baseline.

The same active and reactive flexibility must be sustained over all timesteps. This is the defining feature of the multi-timestep FFOR.

Nodal power injections. At each non-slack MV node, the net nodal injection is obtained from flexible and non-flexible resources:

$$p_{it} = P_{it}^{\mathrm{PV}} + P_{it}^{\mathrm{B}} - P_{it}^{\mathrm{HP}} - P_{it}^{\mathrm{D}} - P_{it}^{\mathrm{EV}} \qquad \forall i \in \mathcal{N} \setminus \{0\},\ t \in \mathcal{T},$$
$$q_{it} = Q_{it}^{\mathrm{PV}} + Q_{it}^{\mathrm{B}} - Q_{it}^{\mathrm{HP}} - Q_{it}^{\mathrm{D}} - Q_{it}^{\mathrm{EV}} \qquad \forall i \in \mathcal{N} \setminus \{0\},\ t \in \mathcal{T}.$$

Linearized power-flow model. To reduce the computational burden, the network is represented through a linearized power-flow model as in [5]. The model uses a sensitivity matrix to relate nodal active and reactive power injections to variations in voltage angles and voltage magnitudes.

The linearization is obtained from a first-order Taylor expansion of the AC power-flow equations around the flat-voltage operating point

$$U_i = 1, \qquad \theta_i = 0 \qquad \forall i \in \mathcal{N}$$

The slack bus, corresponding to the PCC, is fixed as

$$\theta_{0t} = 0, \qquad \Delta U_{0t} = 0 \qquad \forall t \in \mathcal{T}.$$

The nodal active and reactive power balance equations are separated between the slack node and the rest of the distribution network.

For the slack node, the active and reactive power balances include the power exchange variables P_{0t} and Q_{0t}:

$$P_{0t} + \frac{p_{0t}}{S_{\mathrm{base}}} - \sum_{j \in \mathcal{N}} \left(J_{P\theta,0j}\theta_{jt} + J_{PU,0j}\Delta U_{jt} \right) = p_0^{\mathrm{ref}} \qquad \forall t \in \mathcal{T} \tag{1}$$
$$Q_{0t} + \frac{q_{0t}}{S_{\mathrm{base}}} - \sum_{j \in \mathcal{N}} \left(J_{Q\theta,0j}\theta_{jt} + J_{QU,0j}\Delta U_{jt} \right) = q_0^{\mathrm{ref}} \qquad \forall t \in \mathcal{T} \tag{2}$$

For all other non-slack nodes, the power exchange with the external grid is zero, yielding:

$$\frac{p_{it}}{S_{\text{base}}} - \sum_{j\in\mathcal{N}} \left(J_{P\theta,ij}\theta_{jt} + J_{PU,ij}\Delta U_{jt}\right) = p_i^{\text{ref}} \qquad \forall i \in \mathcal{N}\setminus\{0\},\ t\in\mathcal{T} \tag{3}$$

$$\frac{q_{it}}{S_{\text{base}}} - \sum_{j\in\mathcal{N}} \left(J_{Q\theta,ij}\theta_{jt} + J_{QU,ij}\Delta U_{jt}\right) = q_i^{\text{ref}} \qquad \forall i \in \mathcal{N}\setminus\{0\},\ t\in\mathcal{T} \tag{4}$$

The reference active-power injection is zero at the flat-voltage operating point, while the reference reactive-power injection accounts for the line shunt terms:

$$\begin{aligned} p_i^{\text{ref}} &= 0 && \forall i\in\mathcal{N},\\ q_i^{\text{ref}} &= -\sum_{j\in\mathcal{N}_i} b_{ij}^{\text{sh}} && \forall i\in\mathcal{N}. \end{aligned}$$

For each physical line connecting nodes i and j, let:

$$y_{ij} = g_{ij} + jb_{ij} = \frac{1}{r_{ij}+jx_{ij}} \qquad \forall (i,j)\in\mathcal{L}$$

Neglecting shunt conductance, the Jacobian entries $J(i,k)$ relating the injection at node i to the state variable at node k are assembled component-wise as:

$$J_{P\theta}(i,k) = \begin{cases} -\sum_{j\in\mathcal{N}_i} b_{ij}, & i=k\\ b_{ik}, & k\in\mathcal{N}_i\\ 0, & \text{otherwise}\end{cases}$$

$$J_{Q\theta}(i,k) = -J_{PU}(i,k)$$

$$J_{PU}(i,k) = \begin{cases} \sum_{j\in\mathcal{N}_i} g_{ij}, & i=k\\ -g_{ik}, & k\in\mathcal{N}_i\\ 0, & \text{otherwise}\end{cases}$$

$$J_{QU}(i,k) = \begin{cases} -\sum_{j\in\mathcal{N}_i} (2b_{ij}^{\text{sh}} + b_{ij}), & i=k\\ b_{ik}, & k\in\mathcal{N}_i\\ 0, & \text{otherwise}\end{cases}$$

Line active and reactive flows are computed consistently with the same linearization. For the directed flow from bus i to bus j

$$\begin{aligned} P_{ijt}^{\ell} &= -b_{ij}\left(\theta_{it}-\theta_{jt}\right) + g_{ij}\left(\Delta U_{it}-\Delta U_{jt}\right)\\ Q_{ijt}^{\ell} &= -b_{ij}^{\text{sh}}\left(1+2\Delta U_{it}\right) - b_{ij}\left(\Delta U_{it}-\Delta U_{jt}\right) - g_{ij}\left(\theta_{it}-\theta_{jt}\right) \quad \forall (i,j)\in\mathcal{L},\ t\in\mathcal{T}\end{aligned}$$

The reverse flow is obtained by applying the same expressions to the opposite directed line (j,i).

Voltage limits. Voltage magnitudes must remain within the admissible operating range:

$$\underline{U} \le 1 + \Delta U_{it} \le \overline{U} \qquad \forall i\in\mathcal{N},\ t\in\mathcal{T}.$$

Line apparent-power limits. The apparent-power flow on each line is originally constrained by the circular limit

$$\sqrt{\left(P_{ijt}^{\ell}\right)^2 + \left(Q_{ijt}^{\ell}\right)^2} \le \frac{\overline{S}_{ij}}{S_{\text{base}}} \quad \forall (i,j)\in\mathcal{L},\ t\in\mathcal{T}.$$

This circular feasible region is approximated by an inscribed regular polygon with $|\mathcal{K}|$ facets, where

$$a_k = \cos\left(\frac{2\pi k}{|\mathcal{K}|}\right), \quad b_k = \sin\left(\frac{2\pi k}{|\mathcal{K}|}\right), \quad \rho_{\mathcal{K}} = \cos\left(\frac{\pi}{|\mathcal{K}|}\right).$$

The resulting linear constraints are

$$a_k P^{\ell}_{ijt} + b_k Q^{\ell}_{ijt} \leq \rho_{\mathcal{K}} \frac{\overline{S}_{ij}}{S_{\text{base}}} \quad \forall (i,j) \in \mathcal{L},\ t \in \mathcal{T},\ k \in \mathcal{K}.$$

PV flexibility constraints. PV active power can be curtailed between zero and the available generation:

$$0 \leq P^{\text{PV}}_{it} \leq \overline{P}^{\text{PV}}_{it} \qquad \forall i \in \mathcal{P},\ t \in \mathcal{T}.$$

During the flexibility optimization, PV inverters are allowed to provide reactive power within the assumed power-factor capability:

$$-\tan\left(\arccos(\text{pf}^{\text{PV}})\right) P^{\text{PV}}_{it} \leq Q^{\text{PV}}_{it} \leq \tan\left(\arccos(\text{pf}^{\text{PV}})\right) P^{\text{PV}}_{it} \qquad \forall i \in \mathcal{P},\ t \in \mathcal{T}.$$

Heat-pump active and reactive power limits. Heat-pump active-power consumption is bounded by the installed electrical capacity:

$$0 \leq P^{\text{HP}}_{it} \leq \overline{P}^{\text{HP}}_{i} \qquad \forall i \in \mathcal{H},\ t \in \mathcal{T}.$$

The reactive-power consumption of heat pumps is modelled through a fixed power factor:

$$Q^{\text{HP}}_{it} = \tan\left(\arccos(\text{pf}^{\text{HP}})\right) P^{\text{HP}}_{it} \qquad \forall i \in \mathcal{H},\ t \in \mathcal{T}.$$

If heat-pump flexibility is disabled when the baseline heating demand is zero, the following additional constraint is imposed:

$$P^{\text{HP}}_{it} = 0 \qquad \forall i \in \mathcal{H},\ t \in \mathcal{T} : P^{\text{HP,base}}_{it} = 0.$$

Heat-pump thermal dynamics. The indoor temperature evolves according to a first-order aggregated thermal model, extracted from [2]:

$$T^{\text{in}}_{i,t+1} = T^{\text{in}}_{it} + \frac{\Delta t}{C_i}\left(\beta_{it} P^{\text{HP}}_{it} - H_i\left(T^{\text{in}}_{it} - T^{\text{out}}_{it}\right)\right) \qquad \forall i \in \mathcal{H},\ t \in \mathcal{T} \setminus \{\max(\mathcal{T})\}.$$

The initial indoor temperature is fixed to the preferred setpoint:

$$T^{\text{in}}_{i0} = T^{\text{set}} \qquad \forall i \in \mathcal{H}.$$

Thermal comfort is enforced through lower and upper temperature bounds:

$$\underline{T} \leq T^{\text{in}}_{it} \leq \overline{T} \qquad \forall i \in \mathcal{H},\ t \in \mathcal{T}.$$

BESS limits and dynamics. The BESS inverter apparent-power limit is originally defined by the circular constraint

$$\sqrt{\left(P^{\text{B}}_{it}\right)^2 + \left(Q^{\text{B}}_{it}\right)^2} \leq \overline{S}^{\text{B}}_{i} \quad \forall i \in \mathcal{B} \cup \mathcal{B}_{\text{ls}},\ t \in \mathcal{T}.$$

This circular feasible region is approximated by a regular polygon for all storage systems:

$$a_k P^{\text{B}}_{it} + b_k Q^{\text{B}}_{it} \leq \rho_{\mathcal{K}}\, \overline{S}^{\text{B}}_{i} \qquad \forall i \in \mathcal{B} \cup \mathcal{B}_{\text{ls}},\ t \in \mathcal{T},\ k \in \mathcal{K}.$$

The state of charge evolves according to a lossless signed-power model:

$$SOC_{i,t+1} = SOC_{it} - \frac{\Delta t}{E_i^{\mathrm{B}}} P_{it}^{\mathrm{B}} \qquad \forall i \in \mathcal{B} \cup \mathcal{B}_{\mathrm{ls}},\ t \in \mathcal{T} \setminus \{\max(\mathcal{T})\}$$

The initial SOC is fixed prior to the flexibility activation, and the admissible SOC ranges are enforced with distinct bounds to reflect the differing operational limits of distributed and large-scale systems:

$$\begin{aligned} SOC_{i0} &= SOC_i^0 && \forall i \in \mathcal{B} \cup \mathcal{B}_{\mathrm{ls}} \\ \underline{SOC}_i^{\mathrm{dist}} \le SOC_{it} &\le \overline{SOC}_i^{\mathrm{dist}} && \forall i \in \mathcal{B},\ t \in \mathcal{T} \\ \underline{SOC}_i^{\mathrm{ls}} \le SOC_{it} &\le \overline{SOC}_i^{\mathrm{ls}} && \forall i \in \mathcal{B}_{\mathrm{ls}},\ t \in \mathcal{T} \end{aligned}$$

With this convention, $P_{it}^{\mathrm{B}} > 0$ decreases SOC because the battery is discharging into the grid, while $P_{it}^{\mathrm{B}} < 0$ increases SOC because the battery is charging.

Appendix E: Objective Function

The multi-timestep FFOR is constructed by solving the OPF repeatedly along different directions in the active-reactive flexibility plane. For a given direction $d \in \mathcal{D}$, defined by the coefficients (α_d, β_d), the optimization problem is

$$\min \quad \alpha_d P_{\mathrm{PCC}}^{\mathrm{flex}} + \beta_d Q_{\mathrm{PCC}}^{\mathrm{flex}}$$

subject to all constraints in Appendix D. Each optimal solution gives one boundary point of the FFOR,

$$\left(P_{\mathrm{PCC}}^{\mathrm{flex},*}, Q_{\mathrm{PCC}}^{\mathrm{flex},*}\right)$$

Appendix F: Data Sources

The network topology and operational baselines are constructed at an MV-equivalent level. The network data, including the node set $\mathcal{N}$ and line set $\mathcal{L}$, is derived from the geo-referenced synthetic Swiss distribution grid dataset [1]. Low-voltage (LV) quantities downstream of each MV/LV substation are aggregated to the corresponding MV node. The input data projections correspond to the selected scenario years (2030, 2040, or 2050) and are represented at an hourly resolution.

The baseline operation of individual grid assets is defined according to their specific operational logic. Conventional demand (P_{it}^{D}) and EV charging (P_{it}^{EV}) are treated as fixed baseline consumptions, aggregated from LV and MV data in [2], since EVs are not modelled as flexible assets in this study. Their corresponding reactive-power consumptions are denoted by Q_{it}^{D} and Q_{it}^{EV}. Conventional reactive demand is obtained from the bus-specific reactive-to-active demand ratio, whereas EV reactive demand is taken from the corresponding fixed EV profiles.

Similarly, PV generation profiles are aggregated from LV and MV connections. The PV baseline assumes maximum available production ($\overline{P}_{it}^{\mathrm{PV}}$) under the prevailing solar irradiation conditions, with no curtailment and operation at unity power factor:

$$Q_{it}^{\mathrm{PV,base}} = 0.$$

For flexible heating and storage assets, the heat-pump (HP) baseline assumes a stationary indoor temperature equal to the setpoint (T^{set}). Using the building thermal conductance (H_i) and the outdoor temperature (T_{it}^{out}), the baseline thermal power and the corresponding active- and reactive-power consumptions are computed as

$$\begin{aligned} Q_{it}^{\mathrm{heat,base}} &= H_i \max\left(T^{\mathrm{set}} - T_{it}^{\mathrm{out}}, 0\right), \\ P_{it}^{\mathrm{HP,base}} &= \frac{Q_{it}^{\mathrm{heat,base}}}{\beta_{it}}, \\ Q_{it}^{\mathrm{HP,base}} &= \tan(\varphi)\, P_{it}^{\mathrm{HP,base}}, \end{aligned}$$

where $\tan(\varphi)$ is determined from the power factor adopted for the heat pumps.

Conversely, battery energy storage systems (BESS) are assigned zero baseline active and reactive power,

$$P_{it}^{\mathrm{B,base}} = 0, \qquad Q_{it}^{\mathrm{B,base}} = 0,$$

thereby reserving their entire operating range for flexibility provision.

Using the generation-positive convention adopted in the OPF, the net active- and reactive-power baseline injections at node i and time t are therefore given by

$$\begin{aligned} p_{it}^{\mathrm{base}} &= \overline{P}_{it}^{\mathrm{PV}} - P_{it}^{\mathrm{D}} - P_{it}^{\mathrm{EV}} - P_{it}^{\mathrm{HP,base}}, \\ q_{it}^{\mathrm{base}} &= -Q_{it}^{\mathrm{D}} - Q_{it}^{\mathrm{EV}} - Q_{it}^{\mathrm{HP,base}}. \end{aligned}$$

Finally, the time-dependent baseline exchange at the Point of Common Coupling (P_{0t}^{base} and Q_{0t}^{base}) is obtained by solving a baseline network problem using the linearized power-flow model detailed in Appendix D. The model retains line resistance, reactance, and shunt susceptance; however, nonlinear active-power losses are not represented in the first-order linearization.

ChatGPT [7] and OpenAI Codex [8], both powered by GPT-5.5, were used for language revision and coding assistance, respectively. All AI-assisted outputs were reviewed and verified by the authors.

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